



Open Source for the Life Sciences (OS4LS)

Letter of Intent Submission

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Section 1: Applicant Information

Question	Answer
First Name	Jonathan
Last Name	Matthis
Email	info@freemocap.org
Organizational or Institutional Affiliation	FreeMoCap Foundation, Inc.
Country of Residence	United States
Have you previously received funding as a PI under the CZI EOSS program?	No
Organization Name	FreeMoCap Foundation, Inc.
Organization Website	https://freemocap.org
Organization Country	United States
Organization Type	Other Non-profit

Section 2: Proposal Information

Question	Answer
Proposal Title	The Free Motion Capture (FreeMoCap) Project

Short Summary	Life is movement, and the ability to one's body comfortably through the world is central to an individual's experience and quality of life. As such, the ability to record and analyze high quality empirically grounded measurements of full body human movement is essential to our ability to provide the standard of care necessary to ensure health throughout the entire life span. ## Gap: Despite how centrality of human movement to the health across the lifespan, our ability to actually measure high quality, full body human kinematics remains limited. Traditional motion capture methods involve extremely high cost, specialized marker-based motion capture systems that can cost from tens to hundreds of thousands of dollars to build a research quality full body motion capture lab. Even though there are new technologies coming online that will allow us to do marker less motion capture using standard cameras, the development of this technology follows the old pattern, following a reliance on ether proprietary camera hardware or expensive token-based subscriptions (requiring thousands of dollars a year and disincentivizing exploration). ## Goal: A markerless motion capture system capable of providing research grade estimates of full body kinematics at minimal cost could reshape the landscape of human-centered health science and perceptual motor neuroscience. The goal of the FreeMoCap Project is to build exactly that system and support the community of users (students, researchers, athletes, and artists) that has grown around it. ## The Free Motion Capture (FreeMoCap) Project: The goal of the FreeMoCap Project provides a free-and-open-source markerless motion-capture platform that allows researchers to turn any set of cameras into a research grade full body motion capture system. FreeMoCap is the flagship software project of the FreeMoCap Foundation, a certified 501c3 public charity dedicated to supporting development and dissemination of free open source scientific and educational software. FreeMoCap has been validated against to a traditional marker-based motion capture system (Qualisys), and shown to produce clinically meaningful measurements of full body kinematics for gait analysis and functional posture assessment [https://hdl.handle.net/2047/D20859645]. ## Need: Our user base and github start count have both doubled since January 2026 (see #Landscape). As the the community and reach of the FreeMoCap project grows, so does the weight of responsibility for our ability to provide globally spread community of users the quality of motion capture tool that they deserve. ## Deliverables: The resources requested in this proposal will support the core mission of the FreeMoCap Project - to build the best motion capture software the world has ever seen at any price. If successful, we stand reshape the landscape of human-centered research and fundamentally alter our relationship with the movement of our bodies through the world.
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Expected Value	Our goal is to build the best motion capture software the world has ever seen at any price. Success means any researcher, clinician, student, or artist with a handful of cameras can produce research-grade full-body kinematics measurements once locked behind six-figure marker-based labs or expensive subscriptions. A clinician assessing gait, a scientist studying perceptual-motor control, or a coach quantifying athletic movement could capture validated 3D data using hardware they already own, with no per-use fees or vendor lock-in. Upstream, FreeMoCap gives computer-vision and pose-estimation researchers a calibrated multi-camera platform to deploy and validate new tracking models in real capture volumes. Downstream, it produces standardized 3D outputs with quality metadata ready for biomechanics, clinical, and statistical pipelines. For AI enablement and large-scale analysis, we will make FreeMoCap fully programmable and batch-capable, so researchers can process large recording sets, build movement datasets across subjects and conditions, and integrate capture into automated workflows — letting advances in AI-based tracking reach movement researchers everywhere through shared, validated, open infrastructure. Practically speaking, the proposal will support our transition from the current `v1.*` architecture to the far more advanced architecure of FreeMoCap `v2.*` that will allow us to grow into a global scale enterprise software with widespread use and developer support.
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Landscape Analysis	The FreeMoCap Github repository has 9,036 stars. Data collected via optional anonymous user telemetry added in September 2022 represents 51.4k pings from 13.8k unique from over 150 countries [https://freemocap.org/data/]. Our github stars and unique user count have doubled since Jan 2026. The FreeMoCap community Discord server hosts an active community of over 3.9k users, comprising a healthy split of students and experts from a wide variety of backgrounds (including animators, researchers, and roboticists and software engineers) FreeMoCap has been cited in nearly a dozen peer reviewed publications and served as the animation core of several award winning videos games. ## Technological Landscape The gold-standard for motion analysis are proprietary marker-based systems as Vicon, Qualisys, and OptiTrack, or commercial markerless systems such as Theia3D. These tools are costly, closed-source, and difficult to modify, audit, or extend. OpenCap provides an open-source alternative, but its workflow is tied to specific hardware, cloud processing, and a fixed analysis pipeline. Pose2Sim supports functions as a processing workflow rather than an end-to-end capture and analysis system. ## AI Applications FreeMoCap is fundamentally a tool to support astandalone desktop application for high throughput multi-camera AI/ML image analysis. See the FreeMoCap Foundations Track 2 proposal for details on how we plan to expand users ability to build custom AI/ML pipelines.
Project Software Name	FreeMoCap
Project Software Repository URL	https://github.com/freemocap/freemocap
Project Software Website URL	https://freemocap.org

Categories	Scientific computing, Workflows and computational pipelines, Other Clinical Biomechanics
Funding Track	Track 1: Domain-specific Tools
Statement of PI Involvement	Confirm
Policies	Confirm

Thank you for your submission. Please reach out to info@os4science.org with any questions and reference your submission ID as noted above in the email subject.